Multi-Cohort Simulation Study: Shared vs. Study-Specific Factors

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1. Purpose and Background

Prior synthetic validation of the Supervised Bayesian Matrix Factorization (SSBMF) model was carried out only in a **single-cohort** setting. The real application pools several PDAC cohorts that differ both by **platform** (RNA-seq, microarray, proteomics) and by **biology**. In that setting the latent factors fall into two kinds:

- **Shared factors** — gene-expression programs present across *all* studies. These are the cross-cohort biological signal we want to detect and, where they are prognostic, to use for survival prediction.
- **Study-specific factors** — programs private to a *single* study. These absorb platform/batch effects and cohort-private biology that should *not* be mistaken for shared, generalizable signal.

This proposal specifies a simulation that generates data with a **known** shared / study-specific factor structure, so we can test whether the current model **natively** recovers that distinction — the same goal the previously discussed cohort dummy-indicator approach was meant to serve. If the base model already separates shared from study-specific factors, the dummy indicator columns may be unnecessary.

The design incorporates the plan from the last meeting (5/29), with two key elements:

1. **EBMF-grounded ground truth.** Realistic gene programs are taken from an Empirical Bayes Matrix Factorization (`flashier`) fit to the merged real training cohorts. EBMF is also our **unsupervised benchmark** (Section 5).
2. **Shared factors carry survival signal; study-specific factors do not** ($\beta \neq 0$ on shared, $\beta = 0$ on specific). Survival is generated from the shared component only.

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2. Notation

Symbol	Meaning	Shape
C	number of cohorts (studies)	scalar
n_c	patients in cohort c; $n = \sum_c n_c$	scalar
p	number of genes	scalar
K	total number of latent factors	scalar
$\mathcal{S}$	index set of **shared** factors	$\lvert\mathcal{S}\rvert$
$\mathcal{P}_c$	index set of factors **specific** to cohort c	$\lvert\mathcal{P}_c\rvert$
L	patient loadings	$n \times K$
F	gene programs (factors)	$p \times K$
β	log-hazard coefficients	K
Y	observed genomics matrix	$n \times p$
$a_{\text{shared}}, a_{\text{specific}}$	amplitude (signal strength) of shared / specific factors	scalar

Rows of Y (and L) are **stacked by cohort**: rows $1..n_1$ are cohort 1, the next n_2 rows are cohort 2, and so on. Factors partition disjointly: $K = \lvert\mathcal{S}\rvert + \sum_c \lvert\mathcal{P}_c\rvert$. The remaining noise and survival parameters (τ_j, α, λ_0, U_i) are defined inline where they first appear in Section 3.

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3. Data-Generating Process

We *specify* the latent quantities L, F, and β — building in exactly which factors are shared and which are cohort-specific, and which carry survival signal — then *generate* the observed data (Y, time, status) from them. The model is given only the observed data and tasked with recovering the latent structure; because we know the ground truth, every recovery error is directly measurable.

Specified (ground truth): F (gene programs), L (patient loadings with the shared/specific block structure), β (which factors are prognostic). **Generated (model input):** Y (expression matrix), and the survival outcomes time and status. **Model's task:** recover $\hat{F}$, $\hat{L}$, and $\hat{\beta}$ from $(Y, \text{time}, \text{status})$ alone.

The components are described below in the order they are constructed.

3.1 Gene Programs F ($p \times K$) — what each factor "looks like" in gene space

F is specified first because it defines what each latent factor *means* biologically: column k is a p-vector of gene weights describing the k-th gene expression program.

Construction. We use real PDAC gene programs as ground truth rather than arbitrary synthetic ones. Concretely, we fit an Empirical Bayes Matrix Factorization (EBMF; `flashier`, `var_type = 2`, greedy + backfit) to the merged real PDAC training cohorts and extract the resulting gene-loading matrix — $p \times K_{\text{EBMF}}$ columns, each a sparse, data-grounded gene program. These columns are then used as the K columns of F (recycled by index if $K > K_{\text{EBMF}}$).

Two adjustments are made before use:

1. **Unit normalization.** Each column is divided by its L_2 norm so that $\|F_{\cdot k}\| = 1$. This decouples program *shape* from program *amplitude*, letting a separate scalar a_k control how strongly factor k appears in Y.
2. **Non-negativity.** The SSBMF model places a non-negative (`point_exponential`) prior on F — the same NMF-style assumption used at initialization. The EBMF programs are signed; taking their absolute value gives non-negative programs consistent with the model's representational assumption. The EBMF benchmark arm is run non-negative for the same reason.

Each column is then multiplied by an amplitude scalar: a_{shared} for columns assigned to shared factors and a_{specific} for cohort-specific ones. Because $\|F_{\cdot k}\| = 1$, the variance contributed by factor k to Y scales as a_k^2, making the amplitude the direct signal-strength dial.

A synthetic fallback is used when real data are unavailable: each column is a random sparse vector (5% of genes active, equal weights), unit-normalized and sign-adjusted the same way.

3.2 Patient Loadings L ($n \times K$) — who expresses each factor, and how much

L encodes how strongly each patient expresses each gene program. The entry L_{ik} is the *loading* of patient i on factor k: large values mean patient i's expression profile is strongly shaped by program k.

Distribution. Every entry is drawn independently from an Exponential distribution with rate 1:

$$L_{ik} \sim \text{Exponential}(\text{rate} = 1), \qquad \mathbb{E}[L_{ik}] = 1, \quad \mathbb{E}[L_{ik}^2] = 2.$$

The Exponential is non-negative and right-skewed: most patients have a moderate loading near 1, a minority have high loadings, and none are negative. This matches the `point_exponential` prior the model places on L during inference — the ground truth and the prior are on the same scale and in the same family.

Shared vs. specific structure. After drawing all entries, specificity is imposed by zeroing out patients who do not belong to the owning cohort:

- **Shared factor** $k \in \mathcal{S}$: L_{ik} is kept for *every* patient, regardless of cohort. All patients can express a shared program.
- **Cohort-c-specific factor** $k \in \mathcal{P}_c$: L_{ik} is set to **zero** for every patient i not in cohort c. Only patients from cohort c can express this program; it is invisible in all other cohorts.

The resulting L is block-sparse: shared columns are dense (all patients), specific columns are zero everywhere except one cohort block. This is the operational definition of "study-specific" built into the simulation.

3.3 Survival coefficients β (K-vector) — which factors are prognostic

β is **fixed by design** — it is not drawn from a distribution. It encodes which factors the model should identify as prognostic:

- **Shared factors** $k \in \mathcal{S}$: assigned nonzero coefficients, cycling the pattern $(1.5, -1.2, 0.8, -0.5)$ to length $|\mathcal{S}|$. For the primary hybrid scenario ($|\mathcal{S}| = 2$): $\beta = (1.5, -1.2)$, so the first shared factor is positively prognostic (high loading $\rightarrow$ worse outcome), the second is protective.
- **Cohort-specific factors** $k \in \mathcal{P}_c$: $\beta_k = 0$. These factors drive expression patterns within a cohort but carry no survival information. The correct model behaviour is to assign $\hat{\beta}_k \approx 0$ to these factors.

The model receives no information about which β_k are zero — it must infer this from the data. Recovering nonzero β on shared factors (true positives) while keeping $\hat{\beta} \approx 0$ on specific factors (avoiding false positives) is the central evaluation criterion.

3.4 Observed genomics matrix Y ($n \times p$)

Given L, F, we generate what the model will actually observe:

$$Y = LF^\top + E, \qquad E_{ij} \sim \mathcal{N}(0,\, 1/\tau_j), \quad \tau_j \sim \text{Gamma}(\text{shape} = 2,\, \text{rate} = 2).$$

Each gene j has its own noise precision τ_j drawn from a Gamma prior ($\mathbb{E}[\tau_j] = 1$, so typical noise SD ≈ 1). This gives heteroscedastic noise across genes, matching real expression data. Y is column-centered before being passed to the model (standard preprocessing).

At this point the simulation has produced the expression matrix Y — equivalent to what we would observe from an RNA-seq or array experiment — with the latent factor structure (L, F) baked in.

3.5 Observed survival outcomes — time and status

The survival outcomes are generated from the **shared factors only**, via a Weibull proportional-hazards model. Specific factors are deliberately excluded from the survival model ($\beta_k = 0$), so any prognostic signal the model finds on specific factors is a false positive.

Linear predictor (shared factors only):

$$\eta_i = \sum_{k \in \mathcal{S}} L_{ik}\, \beta_k.$$

Event time via Weibull-PH inverse CDF:

$$T_i = \left(\frac{-\log U_i}{\lambda_0 \cdot \exp(\eta_i)} \right)^{1/\alpha}, \quad U_i \sim \text{Uniform}(0, 1),$$

where $\alpha = 1.5$ (Weibull shape) and $\lambda_0 = 0.01$ (baseline scale). This is the standard inversion method: U_i is a uniform random draw, and the formula inverts the Weibull-PH survival function $S(t \mid \eta_i) = \exp(-\lambda_0 t^\alpha e^{\eta_i})$ to produce a random event time that respects the proportional-hazards structure.

Censoring is applied by drawing a censoring time $C_i = s \cdot B_i$ where $B_i \sim \text{Exponential}(1)$ and s is a scale chosen by bisection to achieve approximately 30% censoring:

$$\text{time}_i = \min(T_i, C_i), \qquad \text{status}_i = \mathbf{1}\{T_i \le C_i\}.$$

What the model receives. The model is given only $(Y, \text{time}, \text{status})$ — the same inputs it would receive from a real study. It does not see L, F, or β. Its task is to jointly factorize Y and fit the survival model to recover $\hat{L}$, $\hat{F}$, and $\hat{\beta}$.

> **Anchoring.** Because $\beta \neq 0$ only on shared factors, survival supervision pulls those factors toward the shared component — the model has an outcome-level incentive to represent them accurately. This is the intended mechanism and the key advantage of supervised over unsupervised factorization.

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4. The Three Scenarios

The scenarios differ only in how the K factors are partitioned into shared vs. specific. For the primary setting $C = 2$, $n_c = 150$, $p = 1000$, $K = 6$:

Scenario	$\lvert\mathcal{S}\rvert$ shared ($\beta \neq 0$)	specific per cohort ($\beta = 0$)	Interpretation
All shared	6	0, 0	Studies effectively one cohort
Hybrid	2	2, 2	Both shared and private structure
Nothing shared	0	3, 3	No cross-cohort structure

Nothing-shared is a deliberate false-positive control. With $\mathcal{S} = \emptyset$, $\eta \equiv 0$ and survival carries no factor signal. The correct result is C-index ≈ 0.5 and all $\lvert\hat{\beta}\rvert$ below threshold — the model should *not* fabricate a shared prognostic factor. An optional knob (`specific_prognostic`, default off) can give each cohort a within-cohort prognostic factor if we later want survival signal in this regime.

Worked example — Hybrid, $C = 2$, $n_c = 150$, $p = 1000$, $K = 6$

- **Dimensions:** $n = 300$; Y is 300×1000, L is 300×6, F is 1000×6, β is length 6. Rows 1–150 = cohort 1; 151–300 = cohort 2.
- **Factor labels:** `[shared, shared, specific_1, specific_1, specific_2, specific_2]`.
- **Loadings L:** columns 1–2 dense (all 300 patients). Columns 3–4 zero for rows 151–300 (private to cohort 1). Columns 5–6 zero for rows 1–150 (private to cohort 2). L shows a visible block-sparse pattern.
- **Coefficients:** $\beta = [1.5,\ -1.2,\ 0,\ 0,\ 0,\ 0]$ — only the two shared factors drive survival.
- **What a correct model should find:** factors 1–2 with loadings across *both* cohorts *and* nonzero $\hat{\beta}$; factors 3–6 cohort-private with $\hat{\beta} \approx 0$. The unsupervised EBMF benchmark should recover the same six gene programs but cannot say which two are prognostic.

Secondary sensitivity: a 3-cohort hybrid (2 shared + 1 private per cohort), and a sweep of the shared-to-specific amplitude ratio ($a_{\text{shared}}/a_{\text{specific}}$) within the hybrid regime to stress the case where study-specific signal dominates.

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5. Model Arms

Each simulated dataset (Y, `time`, `status`, `cohort_id`) is fit by **five arms** — four supervised (LB/YFB $\times$ $\pm$cohort indicator) plus the unsupervised EBMF benchmark:

Arm	Model	Cohort indicator
YFB_base	YFB ($\eta = (YF)\beta$) — recommended	none
YFB_cohort	YFB	yes
LB_base	LB ($\eta = L\beta$)	none
LB_cohort	LB	yes
EBMF	unsupervised `flashier` (survival-blind)	—

All supervised arms use a non-negative (`point_exponential`) prior on L and F and a normal prior on β. The cohort indicator, when present, absorbs per-cohort gene-mean differences without entering the survival predictor.

The native-identification test. The headline analysis is the **base** arms (no cohort indicator): can the fitted factors separate into shared vs. study-specific on their own? The $\pm$indicator comparison then quantifies whether the indicator adds anything.

Why EBMF. EBMF is fit to the same Y but is survival-blind, so it has no β. It is the unsupervised gold standard for the factorization itself, answering two questions:

1. Does the supervised model recover the shared/study-specific structure *as well as* EBMF (i.e., supervision does not cost structural fidelity)?
2. Does supervision *additionally* yield correct β — flagging which factors are prognostic — which EBMF fundamentally cannot?

This is the value-add argument: same structural recovery, plus correct survival attribution.

> The EBMF here (fit to each *simulated* Y, scored as a competing method) is distinct from the EBMF in Section 3.2 (fit to *real* data to build ground-truth programs). They do not share a fit.

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6. Evaluation Metrics

For each (scenario $\times$ model arm $\times$ seed) we record:

- **Shared-factor recovery** — mean best |cor| between estimated and true gene programs (F columns, gene space, comparable across LB/YFB/EBMF), over the *true shared* factors.
- **Specific-factor recovery** — same, over the *true study-specific* factors.
- **Specificity-classification accuracy** — label each *estimated* factor shared/specific from its per-cohort loading energy (a factor is called specific if more than 85% of its squared-loading energy concentrates in a single cohort), then compare to the truth. This is the direct test of "does the model natively identify the distinction?"
- **Survival-effect recovery** — $|\hat{\beta}|$ on factors matched to true shared (expect above a small threshold) vs. true specific (expect ≈ 0). Reported as true-positive and **false-positive prognostic rates** (the latter is the anchoring / nothing-shared check).
- **C-index** — orientation-free held-out concordance, $\max(C, 1-C)$.

Comparisons reported: (i) YFB vs. LB; (ii) base vs. +cohort-dummy (Δ in each metric — "are dummies necessary?"); (iii) supervised vs. unsupervised EBMF.

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7. Reference Data-Generating Code

The following is the complete, self-contained DGP (the project implementation wraps this same logic with configuration and seeds). A reader can run it directly given the small `.calibrate` helper.

```r
# Bisection calibration of censoring scale to a target censoring fraction.
.calibrate <- function(ev, base, target = 0.30, n_iter = 40) {
  lo <- 1e-3; hi <- max(ev) * 10
  for (i in seq_len(n_iter)) {
    mid <- sqrt(lo * hi)
    if (mean(base * mid < ev) > target) lo <- mid else hi <- mid
  }
  hi
}

generate_multicohort_data <- function(
    C            = 2L,                 # number of cohorts
    n_per        = rep(150L, C),       # patients per cohort (rows stacked by cohort)
    p            = 1000L,              # genes
    K_shared     = 2L,                 # # shared factors      (beta != 0)
    K_specific   = rep(2L, C),         # # specific factors per cohort (beta = 0)
    F_templates  = NULL,               # p x K_ebmf EBMF programs; NULL -> synthetic sparse
    a_shared     = 1.0,                # amplitude of shared  F columns
    a_specific   = 1.0,                # amplitude of specific F columns
    active_rate  = 0.05,               # synthetic-fallback sparsity
    beta_shared  = NULL,               # length K_shared; NULL -> recycle c(1.5,-1.2,0.8,-0.5)
    offset_sd    = 0.0,                # explicit platform offset SD (0 = off)
```

```r
  shape       = 1.5, scale0 = 0.01, # Weibull-PH baseline (shape = alpha, scale0 = lambda_0)
  target_cens = 0.30,               # target censoring fraction
  seed        = 1L) {

set.seed(seed)
n  <- sum(n_per)
K  <- K_shared + sum(K_specific)
cohort_id <- factor(rep(seq_len(C), times = n_per))

# label + owner per column (owner 0 = shared)
labels <- c(rep("shared", K_shared),
            unlist(lapply(seq_len(C), function(cc) rep(paste0("specific_", cc), K_specific[cc]))))
owner  <- c(rep(0L, K_shared),
            unlist(lapply(seq_len(C), function(cc) rep(cc, K_specific[cc]))))

# L (n x K): Exponential(1), block-zeroed outside owner cohort
L <- matrix(rexp(n * K, rate = 1), n, K)
for (k in seq_len(K)) if (owner[k] != 0L) L[cohort_id != owner[k], k] <- 0

# F (p x K): EBMF templates (unit-norm) or synthetic sparse (unit-norm)
Fm <- matrix(0, p, K)
if (!is.null(F_templates)) {
  idx <- ((seq_len(K) - 1L) %% ncol(F_templates)) + 1L
  Fm  <- F_templates[, idx, drop = FALSE]
  Fm  <- sweep(Fm, 2, sqrt(colSums(Fm^2)) + 1e-10, "/")
} else {
  for (k in seq_len(K)) {
    active <- sample.int(p, max(1L, round(active_rate * p)))
    Fm[active, k] <- 1 / sqrt(length(active))
  }
}
amp <- ifelse(labels == "shared", a_shared, a_specific)   # signal-strength knob
Fm  <- sweep(Fm, 2, amp, "*")

# per-gene Gaussian noise: tau_j ~ Gamma(2,2)
tau <- rgamma(p, shape = 2, rate = 2)
E   <- sweep(matrix(rnorm(n * p), n, p), 2, sqrt(tau), "/")

# optional explicit platform offset (off by default)
offset <- matrix(0, n, p)
if (offset_sd > 0) for (cc in seq_len(C)) {
  ov <- rnorm(p, 0, offset_sd)
  offset[cohort_id == cc, ] <- matrix(ov, sum(cohort_id == cc), p, byrow = TRUE)
}

# assemble + column-center
Y <- L %*% t(Fm) + offset + E
Y <- sweep(Y, 2, colMeans(Y), "-")

# survival from SHARED factors only
if (is.null(beta_shared))
  beta_shared <- if (K_shared > 0) rep_len(c(1.5, -1.2, 0.8, -0.5), K_shared) else numeric(0)
beta <- numeric(K); beta[labels == "shared"] <- beta_shared
eta  <- as.vector(L %*% beta)                  # == 0 in the "nothing shared" scenario

event <- (-log(runif(n)) / (scale0 * exp(eta)))^(1 / shape)
```

```r
    cb    <- rexp(n, rate = 1)
    cs    <- .calibrate(event, cb, target = target_cens)
    ctime <- cb * cs
    time   <- pmin(event, ctime)
    status <- as.integer(event <= ctime)

    list(Y = Y, time = time, status = status, cohort_id = cohort_id,
         L_true = L, F_true = Fm, beta_true = beta,
         factor_labels = labels, factor_owner = owner)
}
```

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8. Initial Results

The design above has been implemented and run end-to-end (3 scenarios $\times$ 5 arms $\times$ 5 seeds). Results are preliminary and may be revised after advisor feedback. Full tables, verification criteria, and the signal-ratio sweep are in the companion results report.

Scenario	Arm	Rec. shared	Rec. specific	Spec. acc.	C-index	β FP
all_shared	YFB_base	0.92	—	1.00	0.87	—
all_shared	YFB_cohort	0.94	—	0.90	0.86	—
all_shared	LB_base	0.53	—	1.00	0.74	—
all_shared	EBMF→Cox	0.82	—	0.90	0.86	—
hybrid	YFB_base	0.90	0.84	0.97	0.81	0.50
hybrid	YFB_cohort	0.92	0.92	1.00	0.81	0.20
hybrid	LB_base	0.72	0.67	0.83	0.76	0.20
hybrid	EBMF→Cox	0.68	0.78	0.80	**0.72**	—
nothing_shared	YFB_base	—	0.89	1.00	0.55	0.03
nothing_shared	YFB_cohort	—	0.91	1.00	0.54	0.03
nothing_shared	LB_base	—	0.78	0.87	0.54	0.07
nothing_shared	EBMF→Cox	—	0.70	0.80	0.55	—

Mean over 5 seeds; LB_cohort omitted for brevity. ***Column definitions: Rec. shared / Rec. specific*** *— gene-program recovery, the mean best |correlation| between estimated and true gene programs over the true shared / study-specific factors (1 = perfect).* ***Spec. acc.*** *— specificity-classification accuracy, the fraction of estimated factors correctly labeled shared vs. study-specific from their per-cohort loading energy.* ***C-index*** *— held-out (external) concordance on the 25% test split (0.5 = chance).* β ***FP*** *— false-positive prognostic rate: fraction of true study-specific factors ($\beta = 0$) wrongly given a survival effect (lower is better).* ***EBMF→Cox*** *is the two-stage baseline: an unsupervised (survival-blind) EBMF factorization followed by a Cox model fit on the EBMF factor scores. It has no β false-positive rate (its factors are not chosen for prognostic relevance); its C-index quantifies how well the "factorize first, predict second" pipeline generalizes, for direct comparison against the joint supervised model. In the hybrid scenario the joint model leads by ≈ 0.09 (0.81 vs. 0.72); in all_shared the two-stage baseline nearly matches (every factor is prognostic), and in nothing_shared both sit at chance.*

The headline findings:

1. **The model natively identifies the shared / study-specific distinction.** In the hybrid scenario, YFB classifies factors as shared vs. specific with accuracy 0.97 (base) to 1.00 (+cohort indicator), purely from per-cohort loading energy — no indicator columns required. This directly answers the central question.
2. **It does not fabricate prognostic signal when none exists.** In nothing_shared ($\eta \equiv 0$ by construction), every arm sits at C-index ≈ 0.54 and the β false-positive rate is 0.03–0.07 — the model does not invent a shared prognostic factor out of purely study-specific structure.
3. **The joint model beats the two-stage EBMF→Cox baseline where it matters most.** In the hybrid scenario — only some factors prognostic, the realistic case — the joint YFB reaches C-index 0.81 vs. 0.72 for "factorize first, fit Cox second" (≈ 0.09 gain). Because EBMF chooses its factors without seeing the outcome, it cannot preferentially sharpen the

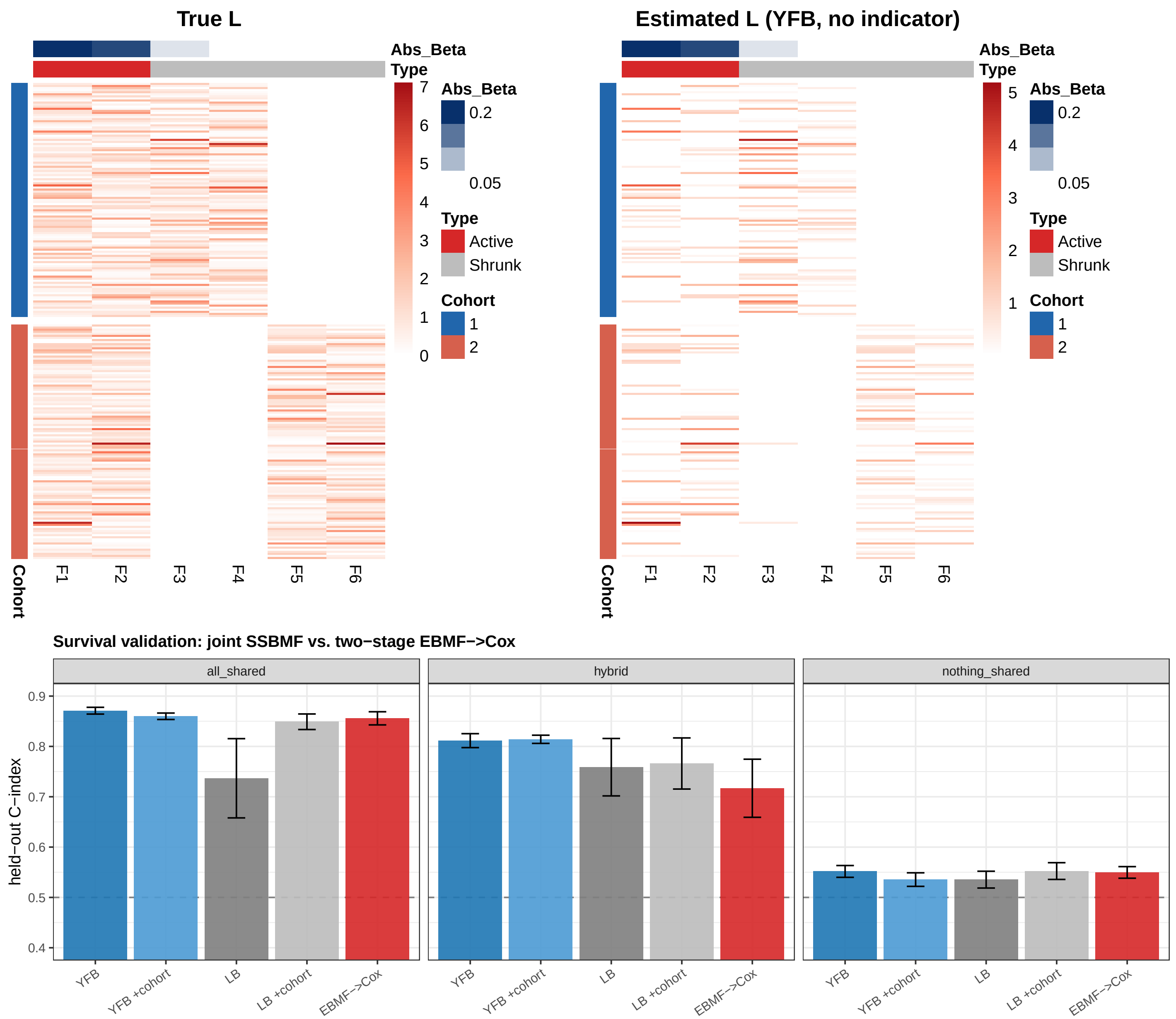

Figure 1: TOP — Patient loading heatmaps for the hybrid scenario (seed 1, training split). Rows = patients grouped by cohort (left sidebar: blue = cohort 1, salmon = cohort 2); columns = factors F1-F6. Top annotation: true factor type (Active = shared/prognostic; Shrunk = study-specific/non-prognostic) and estimated |beta-hat| from YFB. LEFT: true L (ground-truth block-sparse structure); RIGHT: estimated L from YFB recovering the same structure with no indicator columns. BOTTOM — Held-out C-index by scenario and arm (mean +/- SE over 5 seeds; dashed line = chance 0.5). The joint supervised model (YFB) and the two-stage EBMF->Cox baseline differ most in the hybrid scenario: when only some factors are prognostic, jointly modelling genomics and survival recovers the prognostic structure that the survival-blind EBMF factorization cannot preferentially sharpen. In all_shared (every factor prognostic) the two-stage baseline nearly matches; in nothing_shared both sit at chance, as designed.

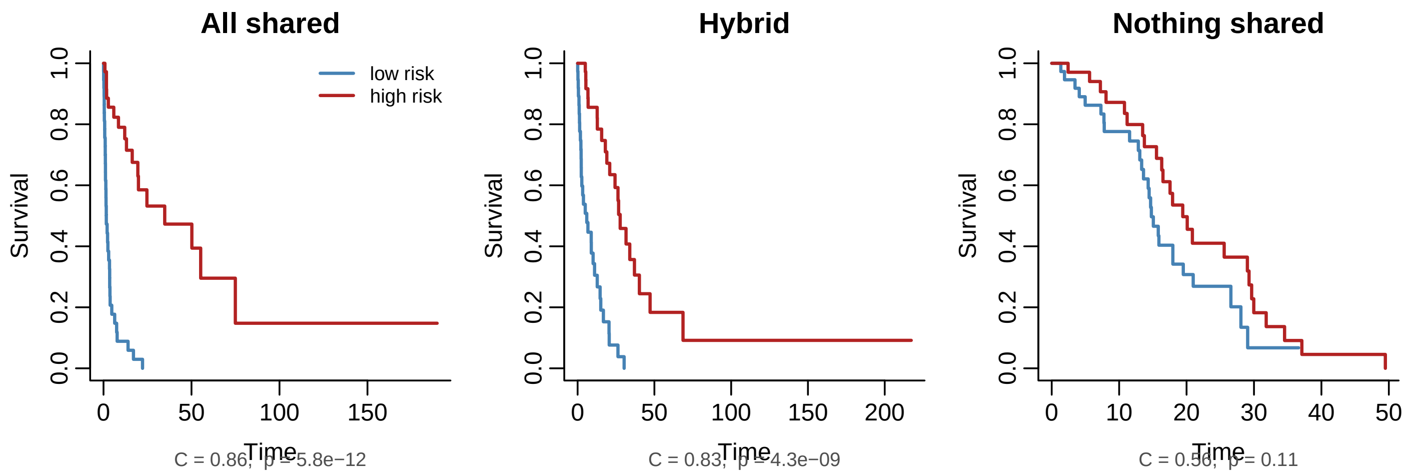

Figure 2: Held-out Kaplan-Meier curves by scenario (YFB_base, seed 1; test split). Patients split at the median predicted risk score. In all_shared every factor is prognostic and separation is strong. In hybrid (the realistic case — shared prognostic factors coexist with study-specific non-prognostic ones) the model cleanly separates risk groups despite never observing the factor labels. In nothing_shared survival is null by construction and the curves correctly overlap (C near 0.5).

prognostic ones; jointly modelling genomics and survival can. In all_shared (every factor prognostic) the two-stage baseline nearly matches (0.86 vs. 0.87), and in nothing_shared both sit at chance — exactly where the gap should and should not appear. This is the core value-add of supervision.

4. **Supervision does not cost structural fidelity.** YFB recovers shared programs at $|\text{cor}| \approx 0.90$–0.94 and specific programs at 0.84–0.92, matching or exceeding the non-negative EBMF benchmark (0.68–0.82).
5. **The cohort indicator helps under genuine heterogeneity, and is redundant otherwise.** In hybrid, adding it halves the β false-positive rate (0.50 $\to$ 0.20) and lifts specificity accuracy to 1.00; in all_shared it adds nothing. This is the quantitative answer to "are the dummy indicators necessary?"
6. **YFB is more reliable than LB**, which collapses to near-zero recovery on some seeds — consistent with YFB being the recommended model.

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9. Open Questions for Discussion

1. **Specificity definition.** We encode study-specific factors via block-zero loadings (a factor is active in exactly one cohort). Is this the structure you intend, or should specific factors also have *non-overlapping gene programs* per cohort?

2. **Nothing-shared survival.** Default leaves survival null in the nothing-shared scenario (false-positive control). Do you prefer instead a within-cohort prognostic specific factor so each cohort has its own survival signal?

3. **Cohort count.** Primary is $C = 2$ (matches the real training data and the fixed-effects rationale); a $C = 3$ hybrid is included as sensitivity. Is 2 sufficient for the methods claim, or should 3 be primary?

4. **Signal ratio.** Default $a_{\text{shared}} = a_{\text{specific}}$. Should the hybrid scenario stress a regime where study-specific variance *dominates* shared variance (the harder, more realistic case)?

5. **Cohort indicator: design decision and remaining false positives.** In the hybrid scenario, adding the cohort dummy indicator halves the β false-positive rate (0.50 $\to$ 0.20), but does not eliminate it — 20% of study-specific factors still receive a spurious prognostic coefficient even with the indicator present.

   The mechanism is confounding: study-specific factors are active in exactly one cohort, and that cohort's patients have different survival trajectories (driven by the shared prognostic factors with different mean loadings across cohorts). Without a cohort label, the model absorbs this cohort-level survival difference by assigning $\hat{\beta}$ to the specific factors. The indicator breaks this confounding, but residual within-cohort correlation (finite sample, $n_c = 150$) accounts for the remaining 0.20.

   Two open sub-questions:

(a) *Should the cohort indicator be part of the default recommended configuration?* The current D4 real-data benchmark achieves C $= 0.636$ **without** the indicator, suggesting confounding is tolerable in practice. But this simulation targets the equal-signal hybrid ($a_{\text{shared}} = a_{\text{specific}}$); whether the PDAC cohorts sit in a harder regime (Q4 above) where the FP problem worsens is unknown.

(b) *What additional approaches could reduce the residual false positives?* Three candidates:

- **Signal-ratio sweep** (planned): stress the specific-dominant regime ($a_{\text{specific}} \gg a_{\text{shared}}$) to quantify how badly the FP rate grows and whether the indicator remains sufficient.
- **Tighter β prior**: a tighter normal prior on β would shrink small spurious coefficients toward zero without the full collapse that the `point_normal` prior causes. This has not been tested in this setting.
- **Cohort-stratified Cox baseline**: replace the shared baseline hazard with cohort-stratified strata, natively absorbing cohort-level survival differences without indicator columns. This is a more substantial model change.

Preliminary sweep results (hybrid, 5 seeds, 5 variance ratios): the FP rate is bounded at 0.20–0.50 across all ratios and does not grow monotonically. The indicator's advantage is largest at equal signal and narrows at high specific-variance ratios. Structural recovery and C-index are stable throughout. Provisional recommendation: include the cohort indicator as default, and use a $|\widehat{\beta}|$ magnitude threshold rather than binary nonzero classification to manage residual false positives.

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10. Recommendation

Recommended model configuration. The YFB variant — linear predictor $\eta = (YF)\beta$, computed from factor *scores* rather than inferred loadings — is the recommended form. On real PDAC data it achieves external concordance $C = 0.636$ across five held-out cohorts (D4 benchmark), exceeding the DeSurv baseline and the LB alternative ($\eta = L\beta$). Key settings that are not negotiable: (i) per-platform z-standardization before pooling — without it, ten of twelve preprocessing configurations collapse $\widehat{\beta} \to 0$; (ii) a `normal` prior on β (the `point_normal` prior causes the same collapse under YFB); and (iii) K selected by cross-validation with the 1-SE rule ($K = 7$ in the D4 benchmark). DeSurv-aligned gene selection (combined-rank, top-3000 per cohort) provides a modest additional gain ($\Delta C \approx +0.012$).

When joint modeling beats the two-stage baseline. The simulation makes the value-add precise: in the *hybrid* scenario — shared prognostic programs coexist with study-specific non-prognostic ones, the realistic structure for multi-cohort PDAC data — the joint YFB model achieves $C = 0.81$ versus 0.72 for EBMF→Cox ($+0.09$). When every factor is prognostic (all-shared) the two-stage baseline nearly matches, and in the null scenario both correctly sit at chance. The gain is therefore concentrated exactly where it matters: when the model must discriminate prognostic shared programs from non-prognostic platform effects *before* fitting survival. EBMF, being survival-blind, cannot preferentially sharpen the prognostic latent dimensions; the CAVI loop does this automatically by coupling the genomics and survival likelihoods.

Why this matters for multi-cohort genomics. Three properties distinguish SSBMF from alternatives:

1. **Joint inference** — genomics and survival are learned simultaneously. The survival likelihood shapes which latent programs are represented, not just which existing programs are associated after the fact.
2. **Interpretable factors** — each column of F is a sparse gene expression program (a GEP), directly analogous to the biological signatures identified by unsupervised NMF or EBMF. The coefficient β_k then quantifies the *prognostic weight* of that program, giving a mechanistic link between molecular programs and outcome.
3. **Cross-cohort generalization** — by explicitly separating shared from study-specific factors at the factorization stage, the model produces risk scores that transfer across heterogeneous platforms and cohorts. The external $C = 0.636$ on five held-out PDAC cohorts, trained on two, demonstrates this. A model that cannot identify and discard platform-specific variation will overfit to cohort identity and generalize poorly.